

REPORT ON LABEL SMOOTHING

1. THE PROBLEM : OVER CONFIDENCE

Before Label Smoothing, neural networks were dangerously Overconfident. Because they were trained to be 100% certain of their predictions, they pushed their internal weights to extreme levels, resulting in models that were totally incapable of expressing doubt, even when they were making mistakes.

HARD TARGET: Standard classification models were trained using absolute, rigid targets. If a picture was a Jean, the network was told the correct answer was $[1.0, 0.0, 0.0, 0.0 \dots]$. It was forced to aim for perfection.

OVER CONFIDENCE: Over Confidence is when a neural network's predicted certainty is significantly higher than its actual accuracy.

THE MECHANICS OF OVERCONFIDENCE:

When a network makes a prediction, it outputs raw scores (logits). To turn those raw scores into percentages, it passes them through a Softmax function.

SOFTMAX FUNCTION: Network calculate their final percentages using a mathematical function called Softmax. The critical flaw is that a Softmax function can only output exactly 1.0 if the internal weights are infinitely large.

Because the network was forced to chase that perfect 1.0 target, it never stops. Even if it guessed a lean with 99% confidence, the loss function will try to make it 100%. The network responds by inflating its weights to massive uncontrollable sizes just to squeeze out fraction of a percent. The network realizes this

and becomes obsessed. It starts inflating its internal weights to massive numbers just to push a 99.1% confidence score up to 99.99%

2. THE DANGERS OF OVERCONFIDENCE;

When the model's weights become massive, it refuses to acknowledge nuance or uncertainty. This causes some major problems.

• CATASTROPHIC REAL-WORLD FAILURES:

If you show an overconfident model a strange-looking shoe it has never seen before, it won't say, "I am 60% sure this is a shoe". It will wildly guess something wrong (like a bag) and be 100% confident about it.

- **RUINED CALIBRATION:** In critical systems (like medical AI or self driving cars). You need to trust the model's self-doubt. If a model says it is 90% sure, it should actually be correct 90 out of 100 times. Over Confident models will say they are 99% sure, but only be right 75% of the time. This means the dashboard metrics are lying to the user.

- **OVER FITTING:** The model stops learning the underlying patterns of the data (generalizing) and instead just memorizes the training data perfectly to achieve that 100% score.

3. MECHANISM OF LABEL SMOOTHING:

PENULTIMATE LAYER: The very last hidden layer right before the final classification output

In Label Smoothing the target is now something like $[0.910, 0.01, 0.01 \dots]$

The network pushes the correct class to a specific finite number.

It forces all incorrect classes to be exactly equal to each other (e.g. exactly 0.01)

If the image is a jacket, the network is mathematically forced to place it in a spot where it is exactly a equal distance away from the Shoe and Jean

Taking example of the fashion MNIST dataset which had 10 outcomes (Shoes, pants etc...)

K (Total classes) : 10

α (Alpha) : 0. (assumed)

tax on correct class = $1 - \alpha = 0.9$

redistribution of the remaining probability

among all classes = $\alpha / K = 0.01$

for the correct class : $0.9 + 0.01 = 0.91$

(taxed on baseline + redistributed value)

for the other 9 classes : $0.01 \times 9 = 0.09$

(redistributed value)

$[1.0, 0.0, 0.0, \dots] \rightarrow \text{Apply } \alpha = 0.1 \rightarrow$

$[0.91, 0.01, 0.01, \dots]$

To achieve this perfect symmetry for every single image, the network physically squashes the data points.

1. TIGHT CLUSTERS: Images of the same class are compressed into incredibly dense, tiny balls.

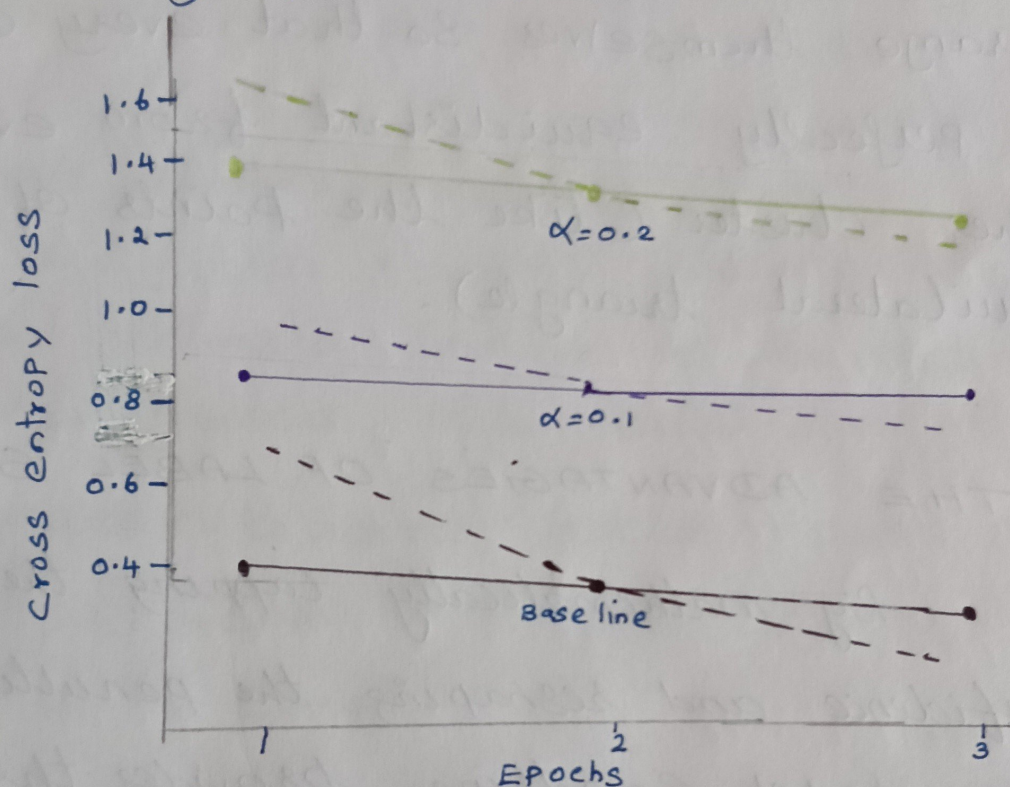
2. EQUIDISTANT SPACING: The clusters arrange themselves so that every cluster is perfectly equidistant from every other cluster (like the points of an equilateral triangle).

4. THE ADVANTAGES OF LABEL SMOOTHING

By mathematically capping the target confidence and reshaping the penultimate layer, Label Smoothing provides three major benefits of the neural network.

- IMPROVED GENERALISATION: Because the model is not forced to chase a 100% target, it doesn't obsess over memorizing the exact pixels of the training set. It learns the general shapes and patterns, which makes it much more accurate when evaluating unseen test data.

- **EXCELLENT CALIBRATION:** The model's confidence scores become highly reliable. If a smoothed model outputs an 85% probability for a T-shirt, it will actually be correct roughly 85% of the time reducing the "Expected calibration Error [ECE]."



LOSS COMPARISON INCREASING SMOOTHING PARAMETER

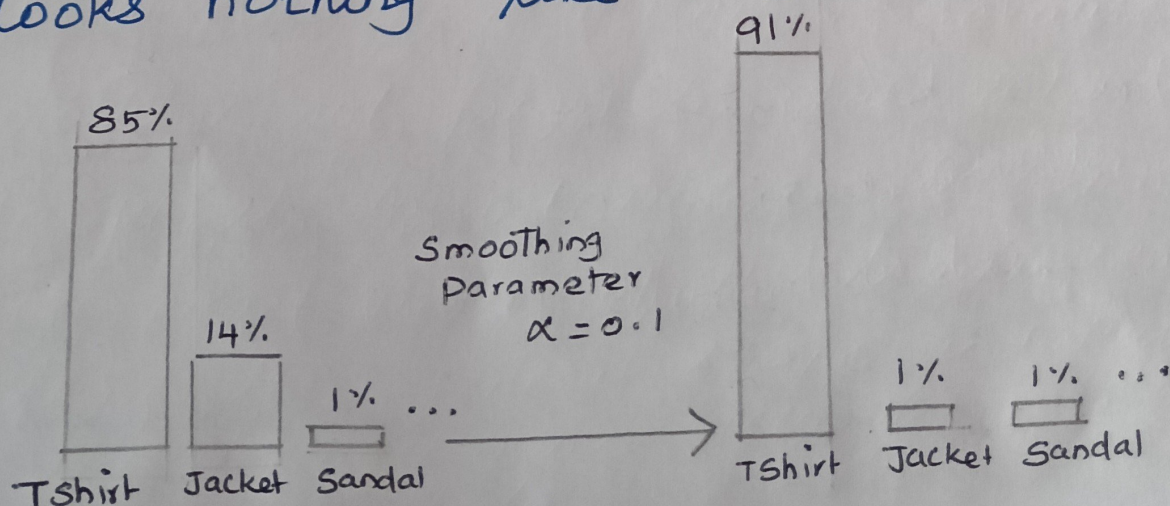
- **TIGHTER FEATURE REPRESENTATIONS:** As proven by the geometric mechanism, Label Smoothing forces images of the same class into incredibly dense, organized clusters, making the final classification boundaries much cleaner.

THE LIMITATIONS: THE KNOWLEDGE DISTILLATION PARADOX;

While label smoothing is a powerful regularization tool, the authors of the paper discovered a massive limitation. It severely damages knowledge distillation.

KNOWLEDGE DISTILLATION: Knowledge distillation is a technique where a massive, highly accurate "Teacher" network is used to train smaller, faster "Student" network. The student learns by looking at the teacher's secondary guesses. often called "Dark Knowledge".

- For example, a normal teacher network might look at a T-shirt and guess. [T-shirt 85%, Jacket 14%, Sandal 1%.]
- That 14% is incredibly valuable information. It teaches the student network that A T-shirt looks a lot like a Jacket, but it looks nothing like a Sandal.



WHY LABEL SMOOTHING RUINS IT: Label Smoothing takes the taxed probability and distributes it equally to all the classes. If we train a teacher model using label smoothing. It is mathematically forced to treat all wrong answers the same.

- It will output (T-shirt 91%, Jacket 1%, Sandal 1%)

By artificially flattening all the secondary guesses. Label smoothing erases the dark knowledge. The teacher model forgets that T-shirts look like jacket. making it a terrible instructor for the student model. Therefore, if a model is going to be used as a teacher later on. Label Smoothing should not be used.

5. POSSIBLE FUTURE IMPROVEMENTS AND EXTENSIONS:

While standard Label Smoothing is highly effective. It applies a rigid. one size fits all mathematical rule. Future research and Extensions can optimize this technique in these specific ways.

• 1. Adaptive Smoothing :

Currently, Label Smoothing distributes the Taxed Probability equally to all incorrect classes. However, in reality, all wrong answers are not equally wrong. For example a T-shirt is visually very similar to a Jacket. but looks nothing like a Sandal. A future improvement would be to distribute the probability based on visual similarity (e.g :- giving 8% to the pullover and 0% to the Sandal) rather than a flat. uniform distribution.

2. INSTANCE-DEPENDENT SMOOTHING:

Standard Label Smoothing uses a fixed alpha (e.g : 0.1) for every single image in the dataset. However, some images are crystal clear. while others are blurry

or ambiguous. A future extension could involve designing a dynamic α that adjusts itself based on the difficulty of understanding the image. The network could learn to apply "Hard Labels" to obvious images and "Soft labels" to confusing ones.

• 3. OVERCOMING THE KNOWLEDGE DISTILLATION BARRIER:

The paper proved the label smoothing destroys the "Dark knowledge" needed for Teacher - Student training. Thus a major area of future research is finding a work around. Extensions could explore "Feature Based Distillation" - where the Student network learns by copying the tight geometric clusters in the Teacher's penultimate layer, rather than relying on the artificially flattened output probabilities.